

Build & Evaluate a Linear Regression Model (House Price Predictor)

Artificial Intelligence & Machine Learning – Task 1

1. Introduction

Objective

The objective of this project is to build and evaluate a Linear Regression model using the California Housing dataset. The project demonstrates the complete Machine Learning workflow including:

- Data loading
- Exploratory Data Analysis (EDA)
- Data visualization
- Feature selection
- Model training
- Prediction
- Model evaluation

The model is trained to predict median house prices using various housing-related features.

2. Dataset Description

The California Housing dataset was loaded using scikit-learn.

Dataset Features

Feature	Description
MedInc	Median income in block group
HouseAge	Median house age
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Block population
AveOccup	Average house occupancy
Latitude	House block latitude

Feature	Description
Longitude	House block longitude
MedHouseVal	Median house value (Target Variable)

Dataset Shape

- Rows: 20640
 - Columns: 9
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3. Exploratory Data Analysis (EDA)

Missing Values

The dataset was checked for missing values using:

```
df.isnull().sum()
```

Observation:

- No missing values were found in the dataset.
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Statistical Summary

The statistical summary was generated using:

```
df.describe()
```

Observations:

- Features such as Population, AveRooms, and AveOccup contain large maximum values.
 - Some features show high variance and skewed distributions.
 - Outliers are present in multiple columns.
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4. Correlation Analysis

A correlation heatmap was plotted to understand relationships between features.

Key Observations

- MedInc shows strong positive correlation with MedHouseVal.
- AveRooms and AveBedrms are highly correlated.
- Latitude and Longitude show strong negative correlation.
- Features like Population and AveOccup have weak correlation with target variable.

Conclusion:

- Median Income (MedInc) is the most influential predictor of house prices.
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5. Data Visualization

Histogram Analysis

Histograms were plotted to understand data distributions.

Observations:

- AveOccup, Population, and AveRooms are heavily right-skewed.
 - Geographic features show clustered distributions.
 - The target variable (MedHouseVal) is moderately skewed.
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Boxplot Analysis

Boxplots were used to detect outliers.

Observations:

- Significant outliers exist in:
 - AveOccup
 - Population
 - AveRooms
 - MedInc
- HouseAge and geographic features contain fewer extreme values.

These outliers may affect Linear Regression model performance.

Scatter Plot Analysis

Scatter plots between MedInc and MedHouseVal show an overall positive linear trend.

Observation:

- Higher median income regions generally have higher house prices.
 - This supports the use of Linear Regression.
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6. Model Building

Feature and Target Selection

```
X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]
```

Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

Linear Regression Model

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

7. Model Evaluation

The model was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Evaluation Metrics

Metric	Value
MAE	0.53
RMSE	.74
R^2 Score	0.57

Interpretation

- Lower MAE and RMSE indicate better prediction accuracy.
 - Higher R^2 score indicates better model performance.
 - The model captures the general trend of house prices reasonably well.
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8. Actual vs Predicted Analysis

An Actual vs Predicted scatter plot was created to compare model predictions.

Observations:

- Predicted values follow the overall trend of actual values.
 - Some deviations exist due to outliers and skewed distributions.
 - Model performance is acceptable for a basic Linear Regression implementation.
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9. Conclusion

This project successfully demonstrated the complete Machine Learning workflow using Linear Regression.

Final Conclusions

- The California Housing dataset was successfully analyzed and visualized.
 - Median Income (MedInc) is the strongest predictor of house prices.
 - Outliers and skewed distributions exist in several features.
 - Linear Regression achieved reasonable prediction performance.
 - The project demonstrates practical implementation of regression analysis using scikit-learn.
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10. Future Improvements

Possible future improvements include:

- Outlier removal
 - Feature engineering
 - Feature scaling
 - Polynomial Regression
 - Random Forest Regression
 - Hyperparameter tuning
 - Cross-validation
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11. Libraries Used

- pandas
- numpy

- matplotlib
 - seaborn
 - scikit-learn
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12. References

- Scikit-learn Documentation
- California Housing Dataset
- Python Data Science Libraries